Name & Surname **Polimi ID**

Exercise 1

A. Please explain when a dependence can be classified as *lexically forward* dependence. (1)

B. What does PCAM Design Methodology mean? (1)

C. Where and under what conditions can the following code be parallelized? Is there a way to write the code for better parallelization if these conditions are met? (2)

```
for (i=0; i<100; i++) {
        for (j=1; j<100; j++)
            a[i][0] += f(a[i-5][j]);
        for (j=0; j<100; j++)
            b[i][j] += f(b[i][j]);
}
```

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Exercise 2

A. Please describe the parallel scan pattern, including assumptions, complexity, and examples. (2)

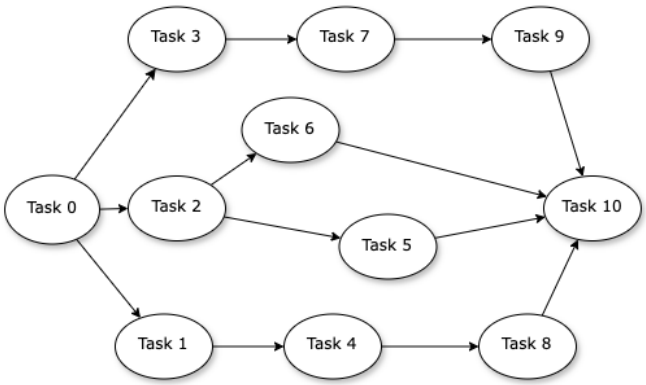
B. Please discuss how the Zip parallel pattern works. How is it implemented (pseudo code required)? (2)

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Exercise 3

A. What are the main differences between Pthreads and OpenMP? (1)

Consider the following task graph:



W(Task 0) = 10
W(Task 1) = 100
W(Task 2) = 10
W(Task 3) = 100
W(Task 4) = 50
W(Task 5) = 50
W(Task 6) = 100
W(Task 7) = 100
W(Task 8) = 50
W(Task 9) = 50
W(Task 10) = 10

B. Calculate Span and Parallelism. If Tasks 0, 2 and 10 represent the thread management overhead, what is the maximum theoretical speedup over a sequential implementation? (1) **This question can be skipped if you got 1 or 3 points in the second challenge.**

C. Write a corresponding OpenMP implementation. (2) **This question can be skipped if you got 2 or 3 points in the second challenge.**

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Exercise 4

A. What is the role of the root parameter in the MPI_Gather function? How does it relate to the sending and receiving buffers? (1)

B. When two MPI processes are running on two different computers, e.g. on two HPC nodes, is it possible that they could end up in a deadlock? (explain the answer) (1)

C. What is an MPI communicator? How and why should I manipulate communicators? Please also provide an example where it is useful. (2)